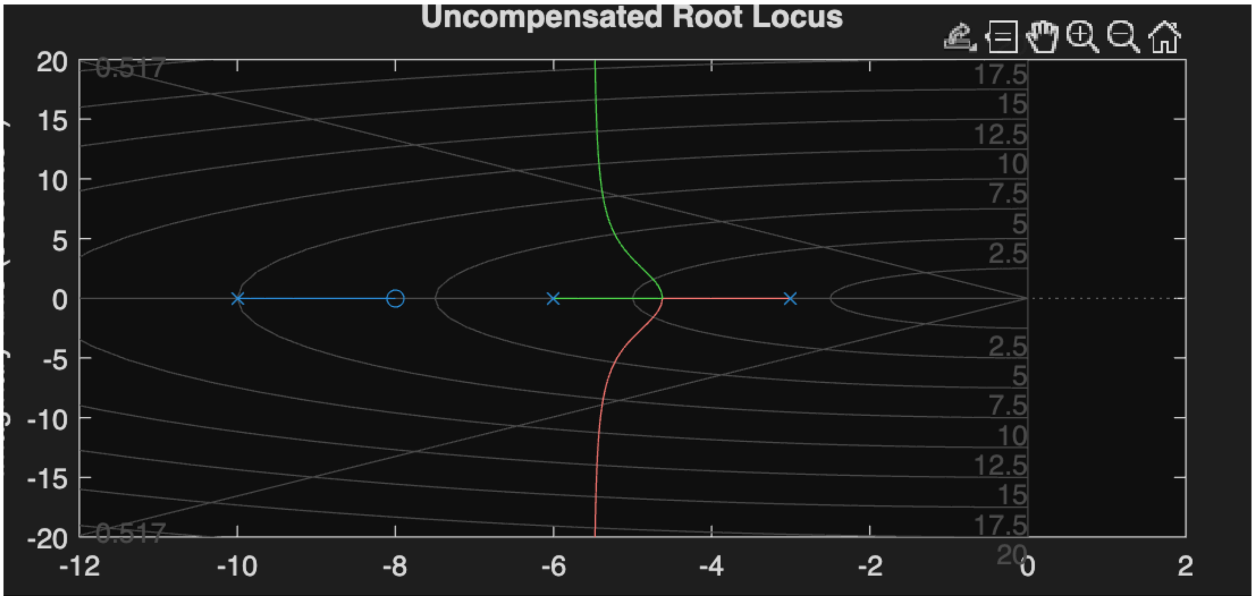
PART A

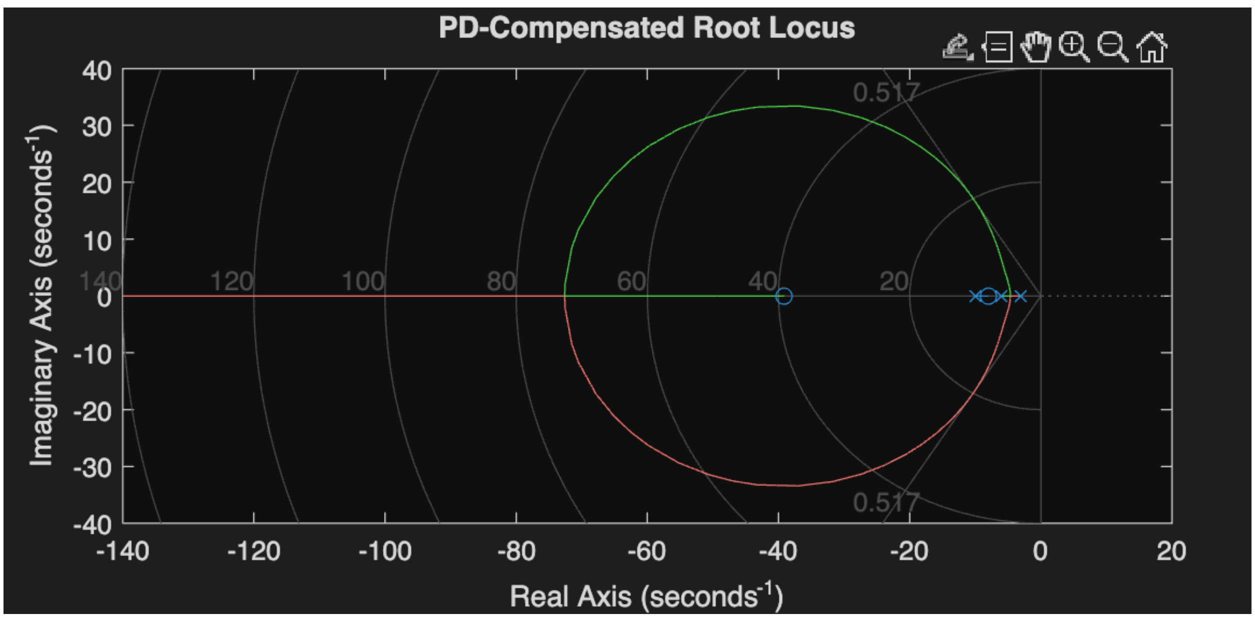
```
clear; clc; close all;
s = tf('s');

%% Plant
Gp = (s+8)/((s+3)*(s+6)*(s+10));

%% Specs
OS = 15;                        % percent overshoot
zeta = -log(OS/100)/sqrt(pi^2 + (log(OS/100))^2);

%% Hand-calculated values
Ku   = 89.1273;                 % uncompensated gain at 15% OS
Tp_u = pi/8.9161;               % old peak time
Tp_n = 0.5*Tp_u;                % new required peak time

zpd  = 39.1367;                 % PD zero location
zpi  = 0.5;                     % PI zero location
Kpd  = 10.5981;                 % PD gain
Kpid = 6.4004;                  % final PID gain

%% Uncompensated closed-loop system
Tu = feedback(Ku*Gp,1);

%% PD-compensated closed-loop system
Gpd = Kpd*(s+zpd);
Tpd = feedback(Gpd*Gp,1);

%% PID-compensated closed-loop system
Gc  = Kpid*(s+zpd)*(s+zpi)/s;
Tpid = feedback(Gc*Gp,1);

%% Display transfer functions
disp('Uncompensated closed-loop TF:');
Tu

disp('PD compensated closed-loop TF:');
Tpd

disp('PID compensated closed-loop TF:');
Tpid

%% Step response info
info_u   = stepinfo(Tu);
```

```
info_pd  = stepinfo(Tpd);
info_pid = stepinfo(Tpid);

disp('Uncompensated step info:');
disp(info_u)

disp('PD compensated step info:');
disp(info_pd)

disp('PID compensated step info:');
disp(info_pid)

%% Root loci
figure;
rlocus(Gp);
sgrid(zeta,[]);
title('Uncompensated Root Locus');

figure;
rlocus((s+zpd)*Gp);
sgrid(zeta,[]);
title('PD-Compensated Root Locus');

figure;
rlocus(((s+zpd)*(s+zpi)/s)*Gp);
sgrid(zeta,[]);
title('PID-Compensated Root Locus');

%% Step responses
figure;
step(Tu, Tpd, Tpid, 5);
grid on;
legend('Uncompensated','PD Compensated','PID Compensated');
title('Step Responses Comparison');
```

```
Uncompensated closed-loop TF:

Tu =

         89.13 s + 713
  ----------------------------
  s^3 + 19 s^2 + 197.1 s + 893

Continuous-time transfer function.
Model Properties
PD compensated closed-loop TF:

Tpd =

     10.6 s^2 + 499.6 s + 3318
  -------------------------------
  s^3 + 29.6 s^2 + 607.6 s + 3498

Continuous-time transfer function.
Model Properties
PID compensated closed-loop TF:

Tpid =

     6.4 s^3 + 304.9 s^2 + 2155 s + 1002
  ----------------------------------------
  s^4 + 25.4 s^3 + 412.9 s^2 + 2335 s + 1002
```

```
Continuous-time transfer function.
Model Properties
Uncompensated step info:
        RiseTime: 0.1578
   TransientTime: 0.7585
    SettlingTime: 0.7585
     SettlingMin: 0.7226
     SettlingMax: 0.9241
       Overshoot: 15.7429
      Undershoot: 0
            Peak: 0.9241
        PeakTime: 0.3507

PD compensated step info:
        RiseTime: 0.0655
   TransientTime: 0.3607
    SettlingTime: 0.3607
     SettlingMin: 0.8793
     SettlingMax: 1.1236
       Overshoot: 18.4563
      Undershoot: 0
            Peak: 1.1236
        PeakTime: 0.1454

PID compensated step info:
        RiseTime: 0.0991
   TransientTime: 2.6227
    SettlingTime: 2.6227
     SettlingMin: 0.9117
     SettlingMax: 1.0992
       Overshoot: 9.9217
      Undershoot: 0
            Peak: 1.0992
        PeakTime: 0.1971
```


Uncompensated Root Locus
0.517
17.5
15
12.5
10
7.5
5
2.5
20
-12
-10
-8
-6
-4
-2
0
2


PD-Compensated Root Locus
0.517
Imaginary Axis (seconds^{-1})
Real Axis (seconds^{-1})
140
120
100
80
60
40
20
-140
-120
-100
-80
-60
-40
-20
0
20

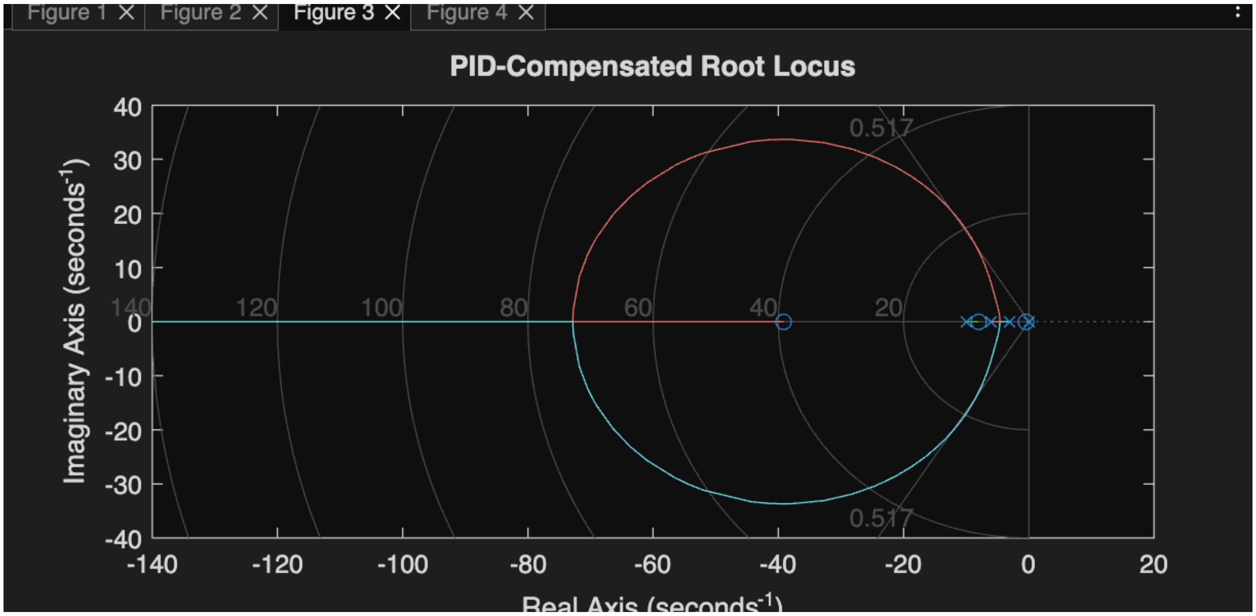
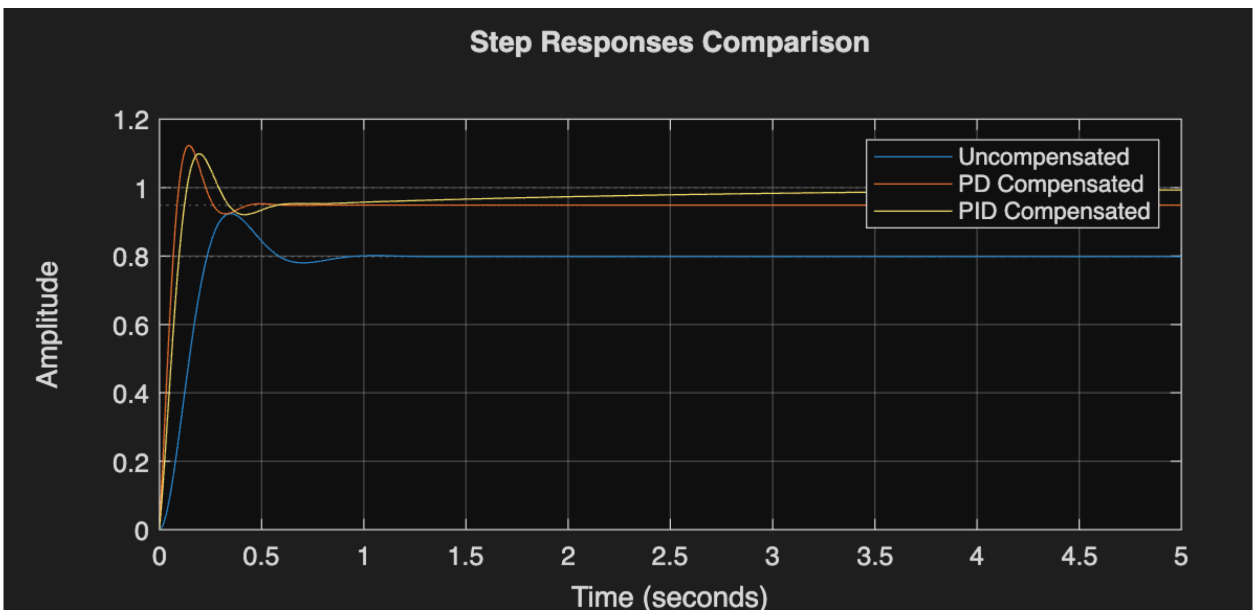


Figure 1
Figure 2
Figure 3
Figure 4
PID-Compensated Root Locus
Imaginary Axis (seconds⁻¹)
Real Axis (seconds⁻¹)
0.517
0.517
140
120
100
80
60
40
20
40
30
20
10
0
-10
-20
-30
-40
-140
-120
-100
-80
-60
-40
-20
0
20

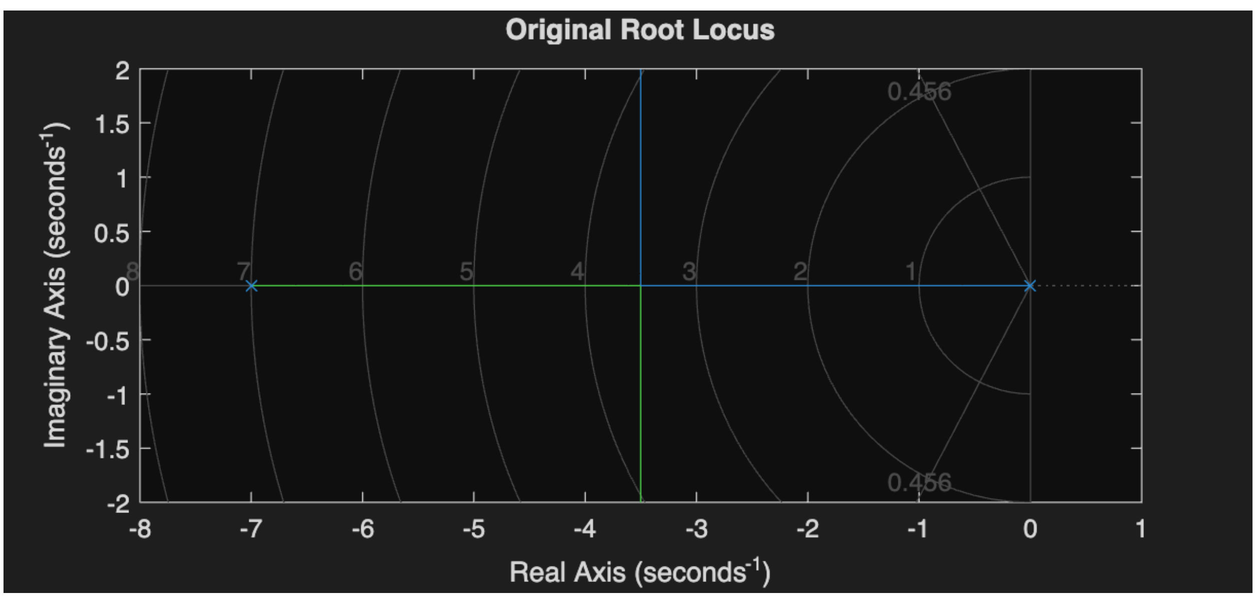

Step Responses Comparison
Amplitude
Time (seconds)
Uncompensated
PD Compensated
PID Compensated
1.2
1
0.8
0.6
0.4
0.2
0
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5

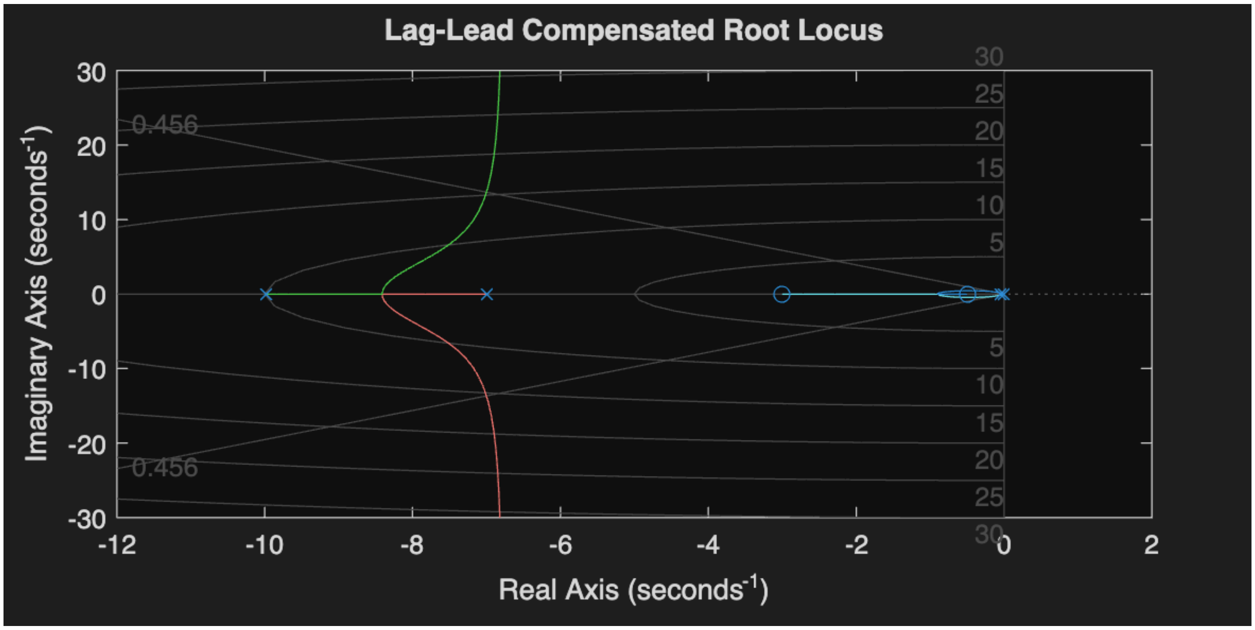
PART B:

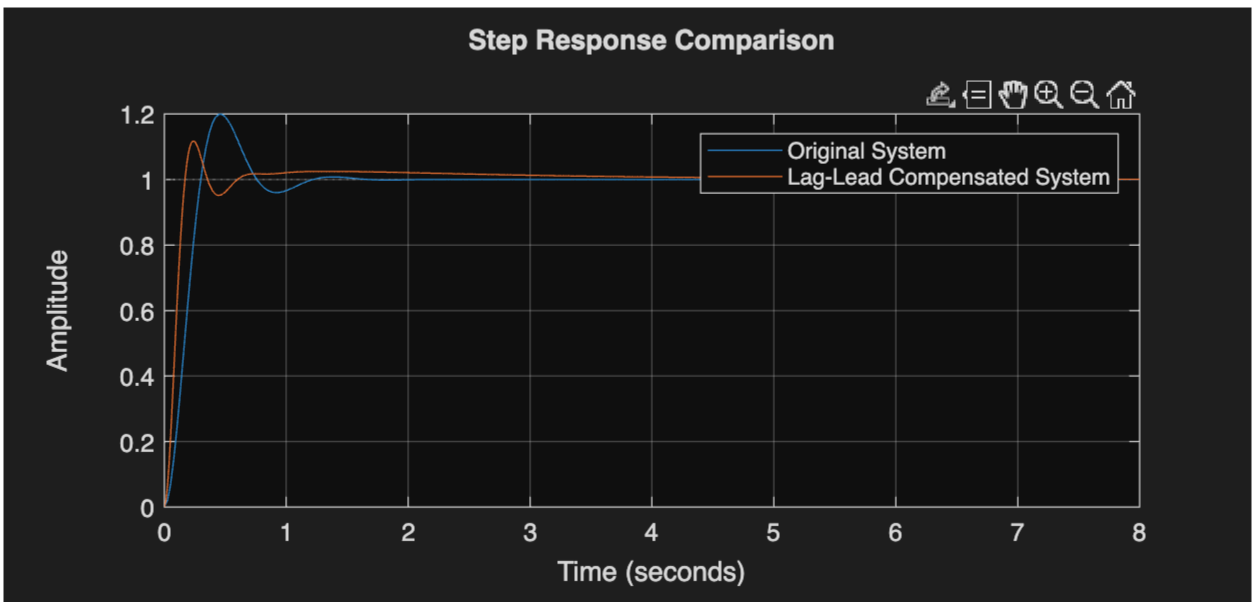
```
clear; clc; close all;
s = tf('s');

%% Original plant
Gp = 1/(s*(s+7));

%% 20 percent overshoot
OS = 20;
zeta = -log(OS/100)/sqrt(pi^2 + (log(OS/100))^2);

%% Original operating point
wn = 7/(2*zeta);
K0 = wn^2;

Ts_old = 4/(zeta*wn);
Kv_old = K0/7;
ess_old = 1/Kv_old;

fprintf('Original gain K = %.4f\n', K0);
fprintf('Original settling time Ts = %.4f s\n', Ts_old);
fprintf('Original ramp steady-state error = %.4f\n', ess_old);

%% New target
Ts_new = 0.5*Ts_old;
ess_new = 0.1*ess_old;

fprintf('Target settling time = %.4f s\n', Ts_new);
fprintf('Target ramp steady-state error = %.4f\n', ess_new);

%% Final lag-lead compensator (hand design)
Kc = 208.7556;
plead = 9.9835;
plag  = 0.05324;
zlead = 3;
zlag  = 0.5;

Gc = Kc*(s+zlead)*(s+zlag)/((s+plead)*(s+plag));

%% Closed-loop system
Tcomp = feedback(Gc*Gp,1);

%% Step info
info_comp = stepinfo(Tcomp);
```

```
disp('Compensated step info:');
disp(info_comp)

%% Velocity error constant
Kv_comp = dcgain(s*Gc*Gp);
ess_comp = 1/Kv_comp;

fprintf('Compensated Kv = %.4f\n', Kv_comp);
fprintf('Compensated ramp steady-state error = %.6f\n', ess_comp);

%% Root locus
figure;
rlocus(Gp);
sgrid(zeta,[]);
title('Original Root Locus');

figure;
rlocus(Gc*Gp/Kc);
sgrid(zeta,[]);
title('Lag-Lead Compensated Root Locus');

%% Step response
figure;
step(feedback(K0*Gp,1), Tcomp, 8);
grid on;
legend('Original System','Lag-Lead Compensated System');
title('Step Response Comparison');
```

```
Original gain K = 58.9253
Original settling time Ts = 1.1429 s
Original ramp steady-state error = 0.1188
Target settling time = 0.5714 s
Target ramp steady-state error = 0.0119
Compensated step info:
         RiseTime: 0.1146
    TransientTime: 2.1190
     SettlingTime: 2.1190
      SettlingMin: 0.9504
      SettlingMax: 1.1179
        Overshoot: 11.7896
       Undershoot: 0
             Peak: 1.1179
         PeakTime: 0.2368

Compensated Kv = 84.1609
Compensated ramp steady-state error = 0.011882
>> Press ⇧⌘P to generate code with Copilot
```




Lag-Lead Compensated Root Locus
Imaginary Axis (seconds^{-1})
Real Axis (seconds^{-1})
0.456
0.456


Step Response Comparison
Original System
Lag-Lead Compensated System
Amplitude
Time (seconds)